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Decreased activity of which of the following enzymes most likely explains the anemia found in this patient?

- ☐ A. δ -aminolevulinic acid dehydratase (19%)
- ☒ B. δ -aminolevulinic acid synthase (60%)
- ☐ C. Cystathionine synthase (5%)
- ☐ D. Glucose-6-phosphate dehydrogenase (11%)
- ☐ E. Pyruvate kinase (2%)

Incorrect

Correct answer
B

60%
Answered correctly

58 secs
Time Spent

This patient with **latent tuberculosis** has **laboratory values** and **bone marrow aspirate** findings consistent with **sideroblastic anemia** due to **isoniazid** use.

Sideroblastic anemia is caused by **impaired heme production** within **red blood cells**. When iron is transported to developing **erythrocytes** that cannot form **heme**, **iron granules** accumulate in **mitochondria** **circumferentially around the nucleus**, forming **ring sideroblasts** that can be seen on **Prussian blue-stained bone marrow aspirate**. Etiologies include **X-linked sideroblastic anemia** (due to an **δ -aminolevulinic acid synthase** mutation), **myelodysplastic syndrome**, **alcohol use disorder**, **copper deficiency**, and certain medications (eg, **isoniazid**, **chloramphenicol**, **linezolid**).

Isoniazid directly inhibits the enzyme **pyridoxine phosphokinase**, which normally converts **pyridoxine** (vitamin B₆) to its active form, **pyridoxal 5' phosphate**. **Pyridoxine** is a **cofactor** for **δ -aminolevulinic acid (ALA) synthase**, the enzyme that catalyzes the rate-limiting step in **heme synthesis**. Inhibition of this enzyme impairs **heme** production and erythroid development to produce a **microcytic anemia**. **Pyridoxine deficiency** can also lead to **seborrheic dermatitis**, **glossitis**, and **peripheral neuropathy**. Therefore, **pyridoxine** is typically prescribed with **isoniazid**.

(Choice A) **δ -aminolevulinic acid dehydratase** is another enzyme involved in **heme synthesis**. **Lead poisoning** can cause **anemia** by directly inhibiting this enzyme. However, **pyridoxine** is not a **cofactor** for this step; therefore, **isoniazid** would not affect its activity.

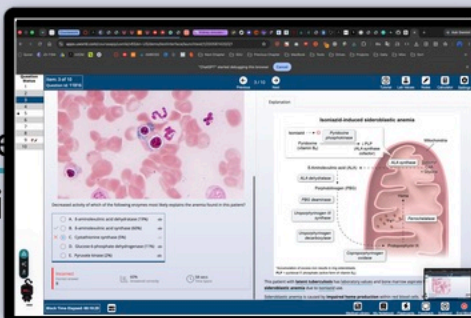
(Choice C) **Cystathionine synthase** catalyzes the formation of **cystathionine** from **homocysteine**. Deficiency results in **homocystinuria**, an **autosomal recessive** disorder characterized by a **marfanoid body habitus** and **hypercoagulability**. Although **cystathionine synthase** is a **pyridoxine-dependent** enzyme, impairment does not result in **sideroblastic anemia**.

(Choice D) **Glucose-6-phosphate dehydrogenase (G6PD) deficiency** results in increased **red blood cell** susceptibility to oxidative stress (eg, infection, medication, fava beans), which triggers **hemolysis**. This results in a **normocytic, normochromic, hemolytic anemia** without **ring sideroblasts**.

(Choice E) **Pyruvate kinase deficiency**, an **autosomal recessive** disorder, results in a **normocytic, normochromic, hemolytic anemia** without **ring sideroblasts**.

Educational objective:

Isoniazid inhibits **pyridoxine phosphokinase**, leading to impaired activation of **pyridoxine**. **Pyridoxine** is a **cofactor** for **δ -aminolevulinic acid synthase**, the enzyme that catalyzes the rate-limiting step in **heme synthesis**. Inhibition of this step can result in **sideroblastic anemia**.



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☐ Mark Question

A 13-year-old girl is brought to the clinic by her mother for a yearly physical examination. The patient feels well but is worried that she has not yet started puberty. Temperature is 36.7 C (98 F), blood pressure is 152/91 mm Hg, pulse is 75/min, and respirations are 18/min. Physical examination is significant for a lack of secondary sexual characteristics; a blind vagina is noted on pelvic examination. Laboratory studies reveal hypokalemia and low testosterone and estradiol levels. Cytogenetic analysis shows a 46,XY karyotype. This patient most likely has deficiency of which of the following enzymes?

☐ A. 5 alpha-reductase

☒ B. 11 beta-hydroxylase

☐ C. 17 alpha-hydroxylase

☐ D. 20,22-desmolase

☐ E. 21-hydroxylase

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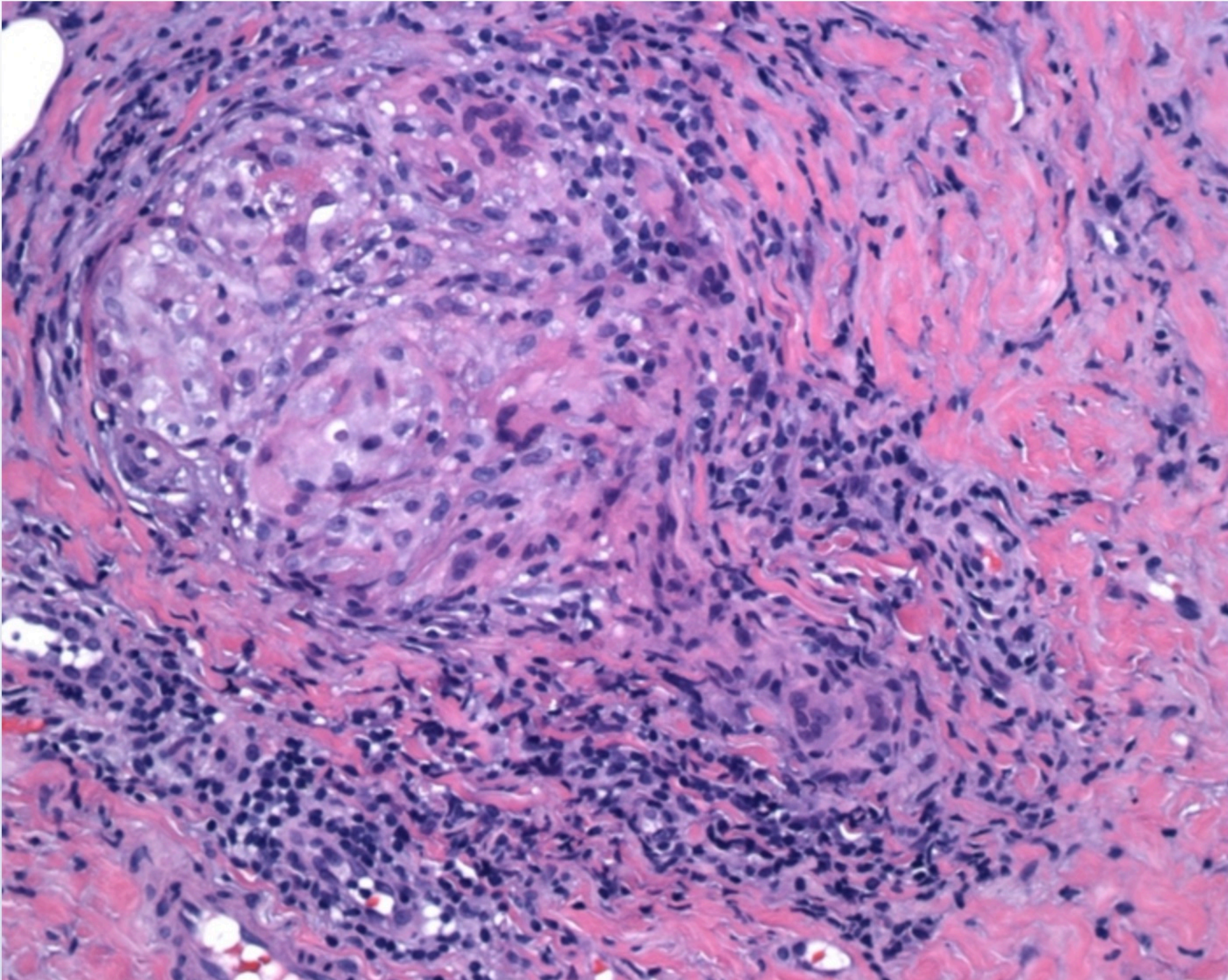
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Mark Question

A 19-year-old man comes to the office due to a painful subcutaneous forearm nodule. Two weeks earlier, he received a laceration on his left forearm while playing soccer and had sutures placed. The lesion is biopsied, and results are shown in the image below:



Which of the following is most likely responsible for this patient's condition?

☒ A. Excess granulation tissue formation

☐ B. Malignant tumor formation

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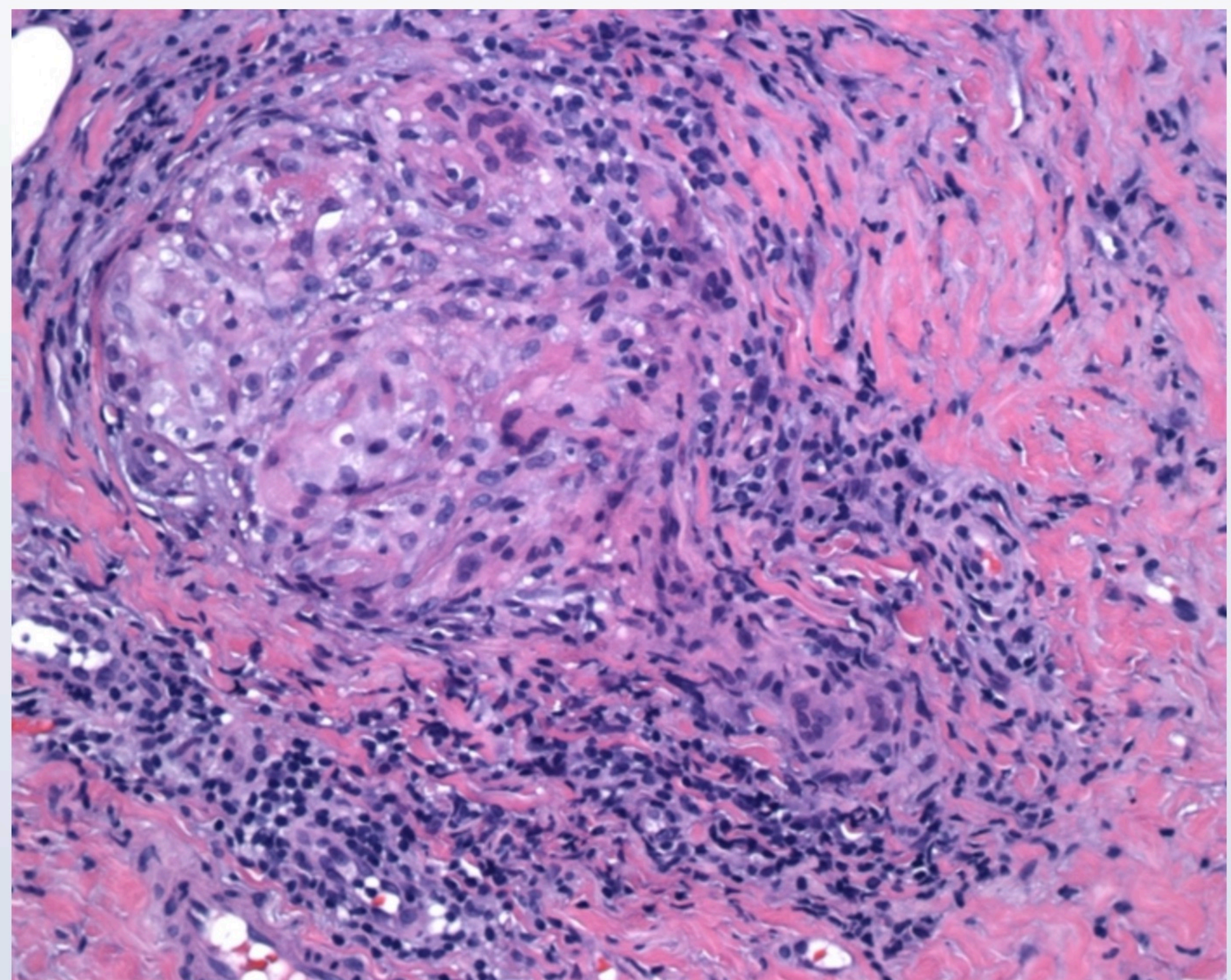
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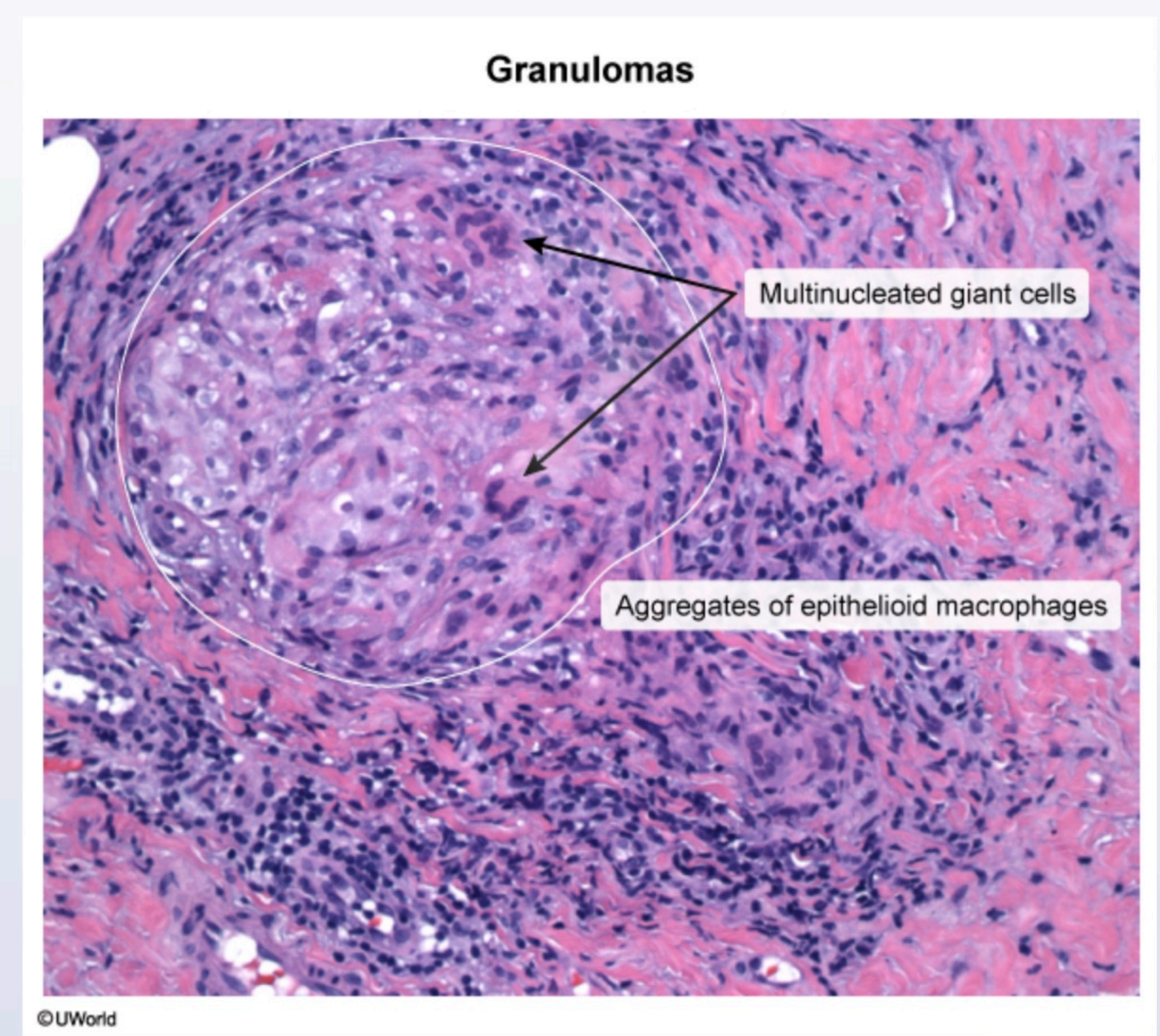
A 19-year-old man comes to the office due to a painful subcutaneous forearm nodule. Two weeks earlier, he received a laceration on his left forearm while playing soccer and had sutures placed. The lesion is biopsied, and results are shown in the image below:



Which of the following is most likely responsible for this patient's condition?

- ☐ A. Excess granulation tissue formation (26%)
- ☒ B. Malignant transformation (0%)

Explanation



Foreign bodies (eg, retained sutures) can elicit a granulomatous response, seen clinically as a tender, erythematous, brown or purple papule, nodule, or plaque. **Granulomas** are characterized microscopically by aggregates of **activated macrophages** that assume an epithelioid appearance. They are rimmed by **lymphocytes** that synthesize the **cytokines** responsible for **granuloma** maintenance. **Multinucleated giant cells**, derived from the fusion of several **macrophages**, can also be found in **granulomas**. **Granuloma** formation is a complex process that occurs over several **days to weeks** in response to a variety of stimuli.



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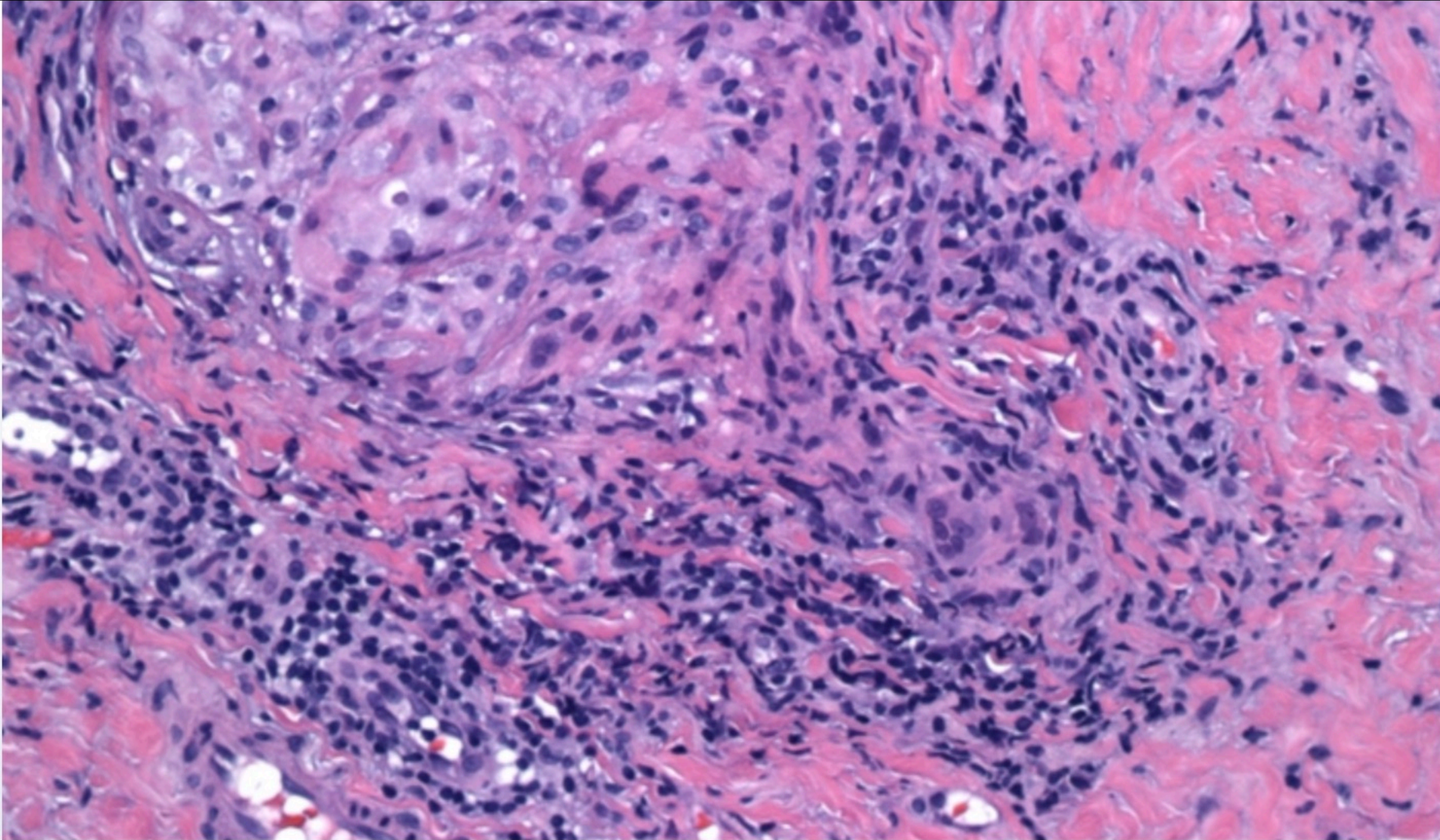
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Which of the following is most likely responsible for this patient's condition?

☐ A. Excess granulation tissue formation (26%)

☒ B. Malignant transformation (0%)

☐ C. Normal wound remodeling (4%)

☐ D. Persistent neutrophil activity (13%)

☐ E. Reaction to a foreign body (54%)

Incorrect

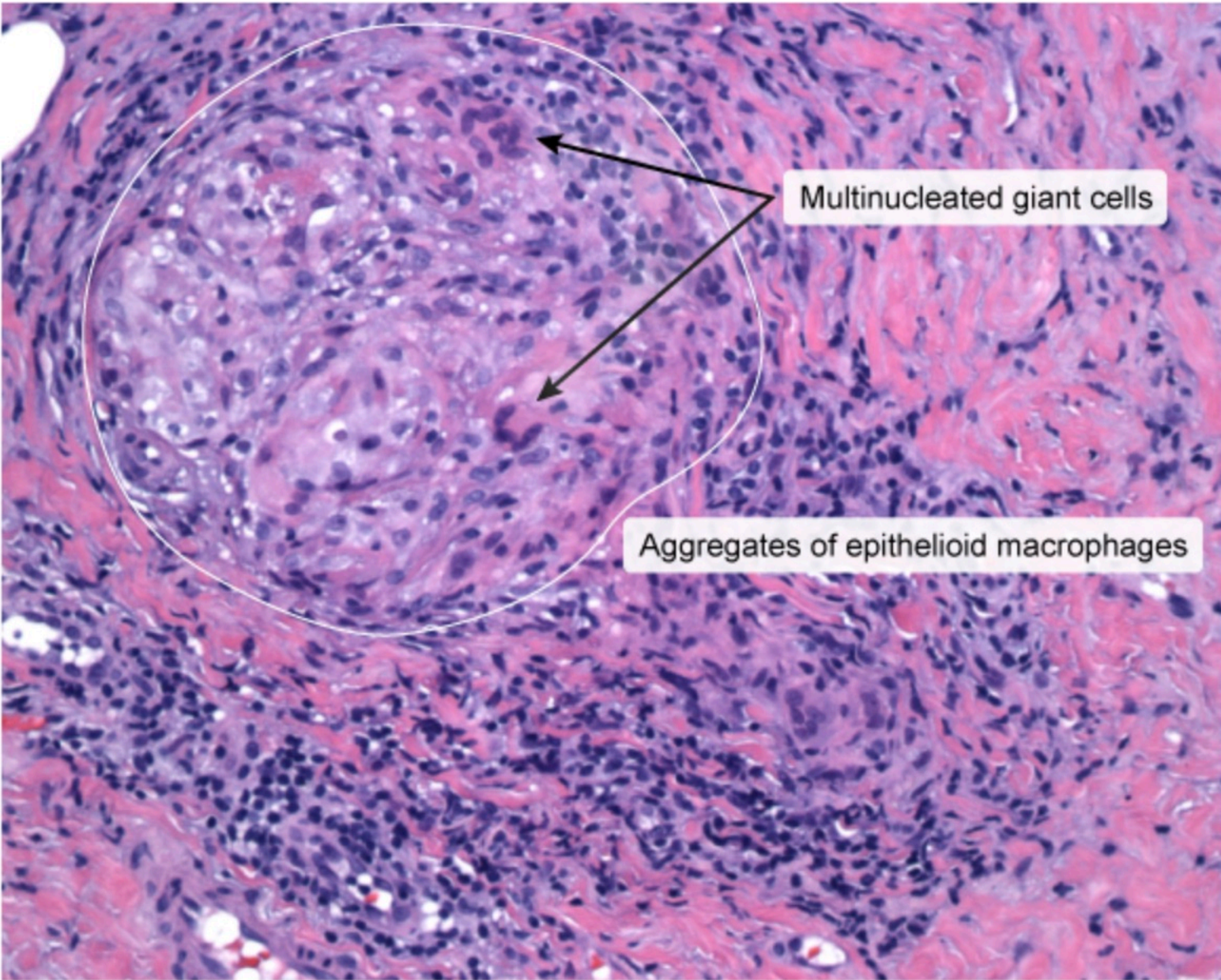
Correct answer E

54% Answered correctly

08 secs Time Spent

Explanation

Granulomas

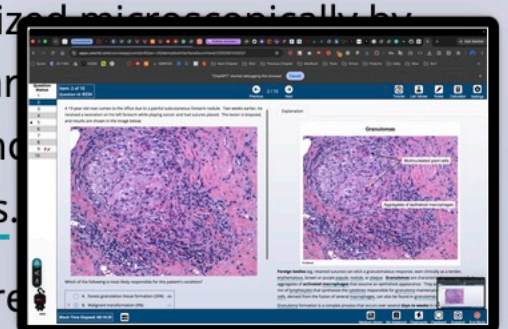


Multinucleated giant cells

Aggregates of epithelioid macrophages

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Foreign bodies (eg, retained sutures) can elicit a granulomatous response, seen clinically as a tender, erythematous, brown or purple papule, nodule, or plaque. **Granulomas** are characterized microscopically by aggregates of **activated macrophages** that assume an epithelioid appearance. They are rimmed by **lymphocytes** that synthesize the **cytokines** responsible for **granuloma** maintenance. **Cells**, derived from the fusion of several **macrophages**, can also be found in **granulomas**. **Granuloma** formation is a complex process that occurs over several **days to weeks** in response to a variety of stimuli.



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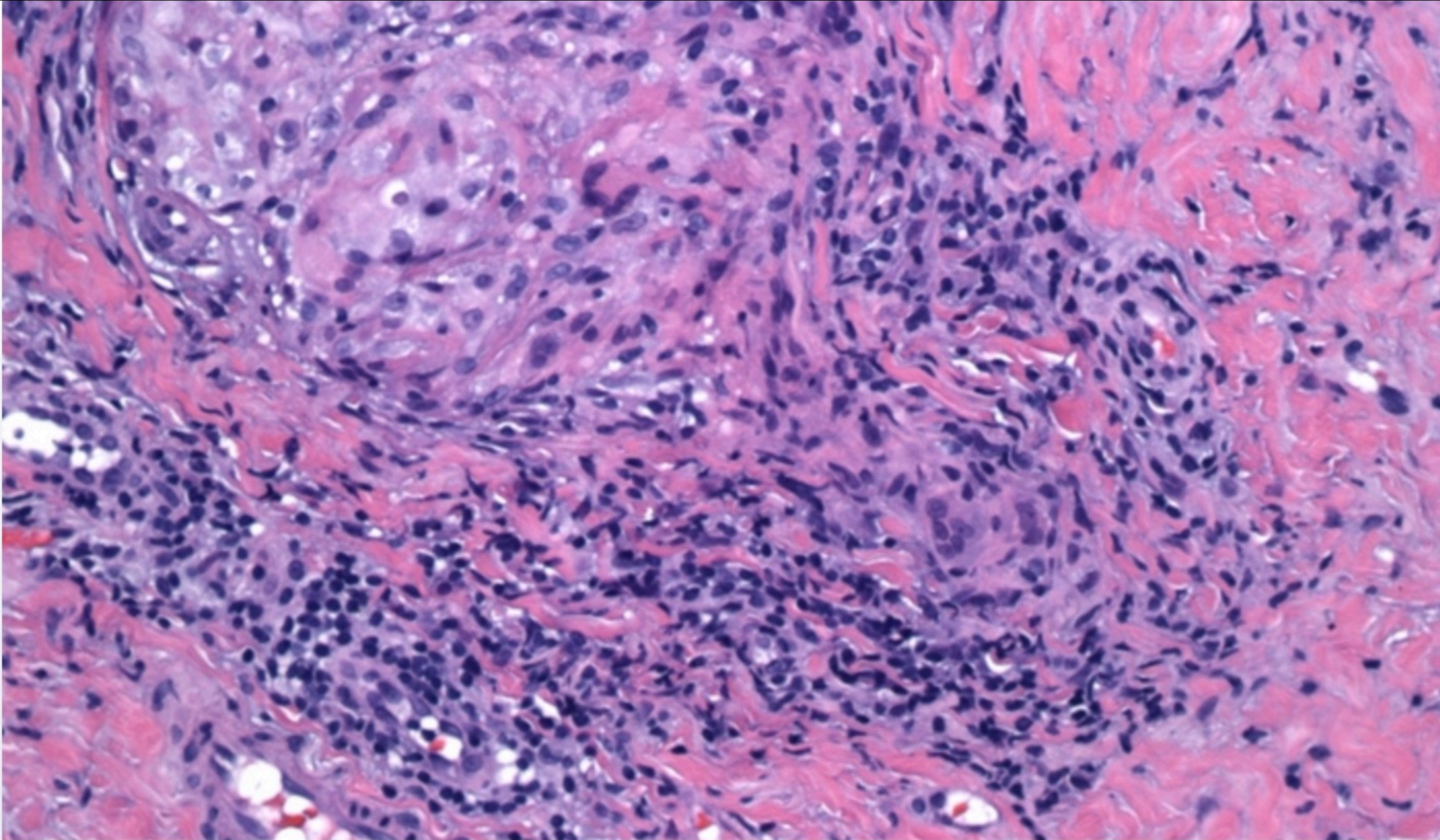
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Which of the following is most likely responsible for this patient's condition?

☐

A. Excess granulation tissue formation (26%)

ab

☒

B. Malignant transformation (0%)

ab

☐

C. Normal wound remodeling (4%)

ab

☐

D. Persistent neutrophil activity (13%)

ab

☒

E. Reaction to a foreign body (54%)

ab

Incorrect

Correct answer E

54% Answered correctly

08 secs Time Spent

Foreign bodies (eg, retained sutures) can elicit a granulomatous response, seen clinically as a tender, erythematous, brown or purple papule, nodule, or plaque. **Granulomas** are characterized microscopically by aggregates of **activated macrophages** that assume an epithelioid appearance. They are often surrounded by a rim of **lymphocytes** that synthesize the **cytokines** responsible for **granuloma** maintenance. **Multinucleated giant cells**, derived from the fusion of several **macrophages**, can also be found in **granulomas**.

Granuloma formation is a complex process that occurs over several **days to weeks** in response to an **antigen** that cannot be eradicated by more acute inflammatory mechanisms (eg, killing by **neutrophils**). As infiltrating **macrophages** engulf the **antigenic debris**, material resistant to digestion persists in the **cytoplasm** and causes **macrophage** activation. This results in further **macrophage** maturation and formation of a **granuloma**, which **walls off** the offending agent.

(Choice A) Excess **granulation tissue** formation (hypergranulation) usually occurs in a wound left to **heal by secondary intention**. Light microscopy would show proliferation of **fibroblasts**, **collagen** deposition, inflammatory cells, and numerous capillaries.

(Choice B) **Marjolin ulcer** is an aggressive, ulcerating **squamous cell carcinoma** that presents in an area of previously traumatized, chronically inflamed, or scarred skin. Malignant transformation often occurs long after the initial trauma, usually >10 years later. **Histology** would show **dysplastic keratinocytes** and keratinization.

(Choice C) The remodeling phase of wound healing is characterized by **fibrosis** (scar formation) and starts the second week after injury. During this phase, active **fibroblasts** synthesize **collagen**, **elastin**, and other components of the **connective tissue** matrix.

(Choice D) The inflammatory phase of wound healing consists of the movement of **neutrophils** (within the first 24 hours) and **macrophages** (within 2-3 days) to the damaged area, with **phagocytosis** of **necrotic debris** and bacterial contaminants. Persistent **neutrophil** activity in a wound would result in an **abscess** and/or **pus** drainage rather than **granuloma** formation.

Educational objective:
Granulomatous inflammation is a form of **chronic inflammation** characterized by aggregate **macrophages** that assume an epithelioid appearance. Persistent **granulomatous inflammation** and **fibrosis** can cause organ dysfunction, which is seen in a number of **granulomatous diseases**.

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Mark Question

A 40-year-old woman comes to the office with a 3-month history of progressively limited physical activity due to fatigue. The patient says, "I could barely walk from my car to the office." Medical history is significant for a positive tuberculin skin test 7 months ago with a normal chest radiograph. She has been compliant with the prescribed treatment despite its bitter taste. Physical examination shows a tired-appearing woman with conjunctival and palmar pallor. Complete blood count is as follows:

Hemoglobin	9 g/dL
Hematocrit	28%
Mean corpuscular volume	72 fL

A bone marrow aspirate representative of this patient's disease process is shown below (Prussian blue stain).

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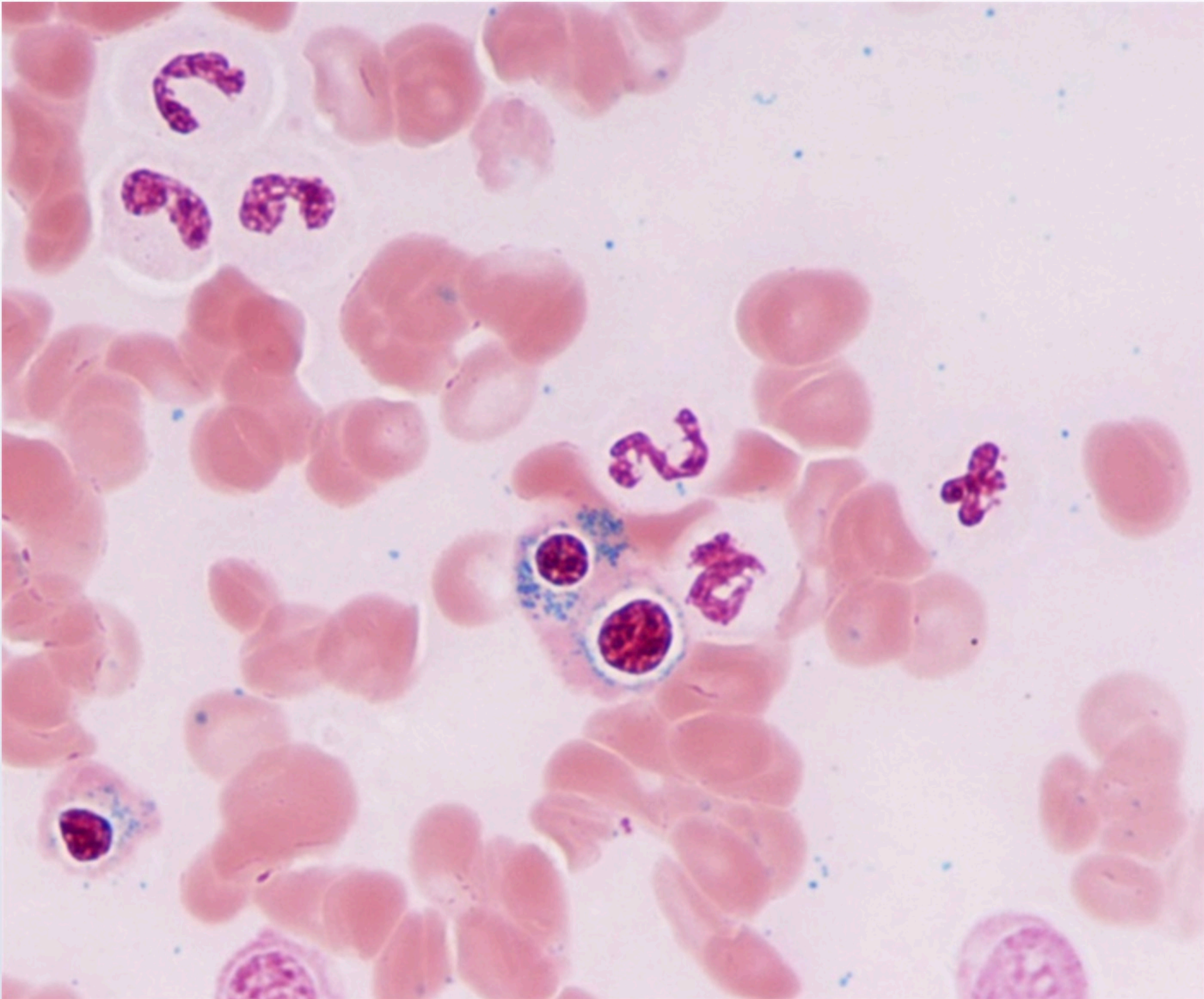
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A bone marrow aspirate representative of this patient's disease process is shown below (Prussian blue stain).



Decreased activity of which of the following enzymes most likely explains the anemia found in this patient?

☐ A. δ -aminolevulinic acid dehydratase

☐ B. δ -aminolevulinic acid synthase

☒ C. Cystathionine synthase

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Decreased activity of which of the following enzymes most likely explains the anemia found in this patient?

- ☐ A. δ -aminolevulinic acid dehydratase (19%)
- ☒ B. δ -aminolevulinic acid synthase (60%)
- ☐ C. Cystathionine synthase (5%)
- ☐ D. Glucose-6-phosphate dehydrogenase (11%)
- ☐ E. Pyruvate kinase (2%)

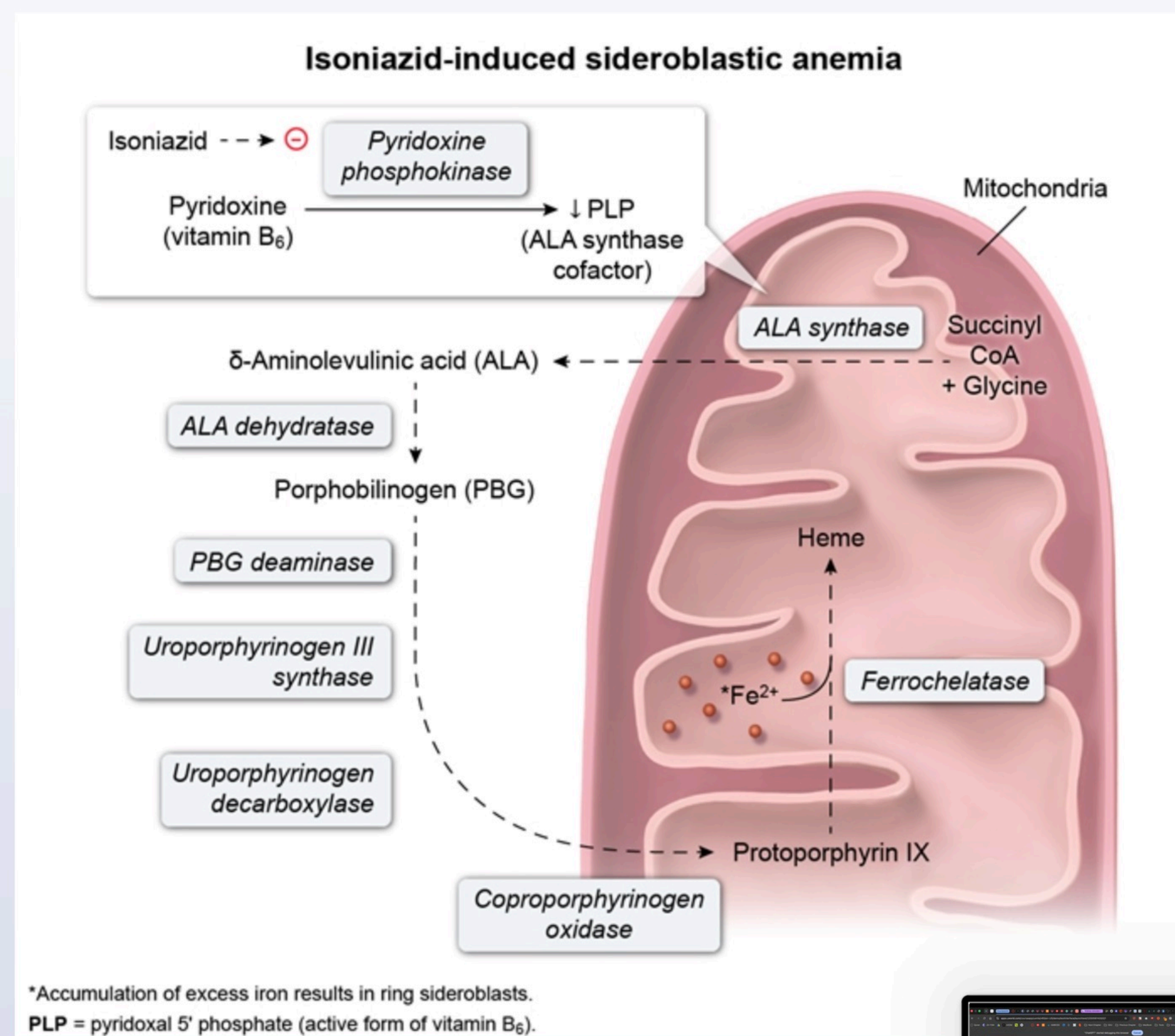
Incorrect

Correct answer B

60%
Answered correctly

58 secs
Time Spent

Explanation



This patient with **latent tuberculosis** has laboratory values and bone marrow aspirate findings consistent with **sideroblastic anemia** due to **isoniazid** use.

Sideroblastic anemia is caused by **impaired heme production** within red blood cells. W

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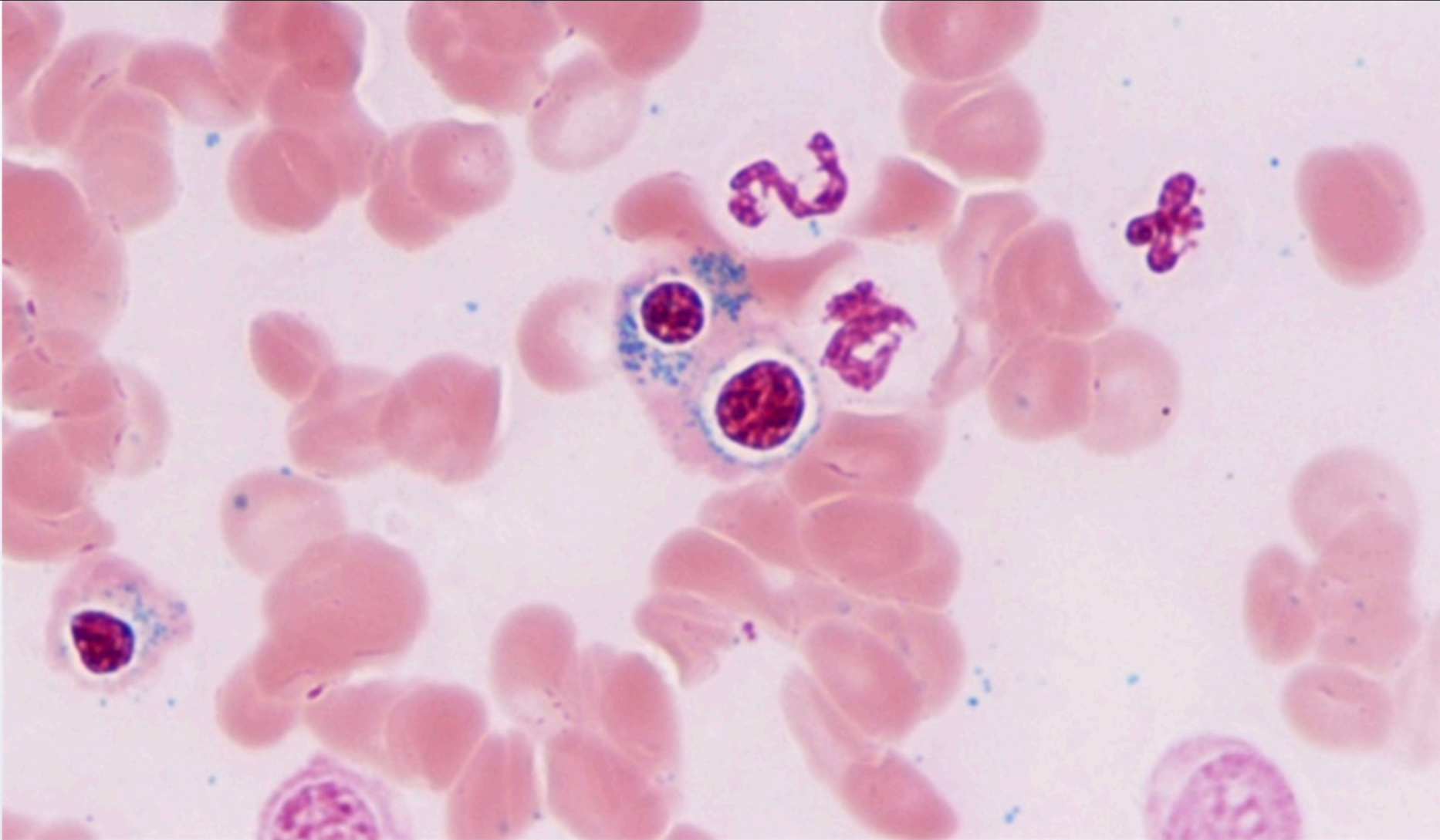
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Decreased activity of which of the following enzymes most likely explains the anemia found in this patient?

☐

A. δ -aminolevulinic acid dehydratase (19%)

ab

☒

B. δ -aminolevulinic acid synthase (60%)

ab

☐

C. Cystathionine synthase (5%)

ab

☐

D. Glucose-6-phosphate dehydrogenase (11%)

ab

☐

E. Pyruvate kinase (2%)

ab

Incorrect

Correct answer B

60% Answered correctly

58 secs Time Spent

Severe isoniazid related sideroblastic anemia.

Biochemistry

Hematology & Oncology

Sideroblastic anemia

Subject

System

Topic

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Severe isoniazid related sideroblastic anemia

Rein Jan Piso¹, Kveti Kriz, Marie-Claire Desax

Affiliations + expand
PMID: 22184524 PMCID: [PMC3238472](#) DOI: [10.4081/hr.2011.e2](#)

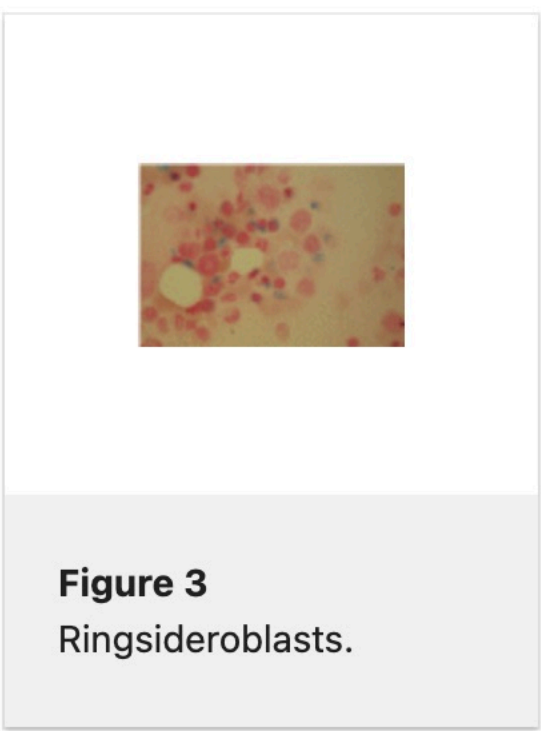
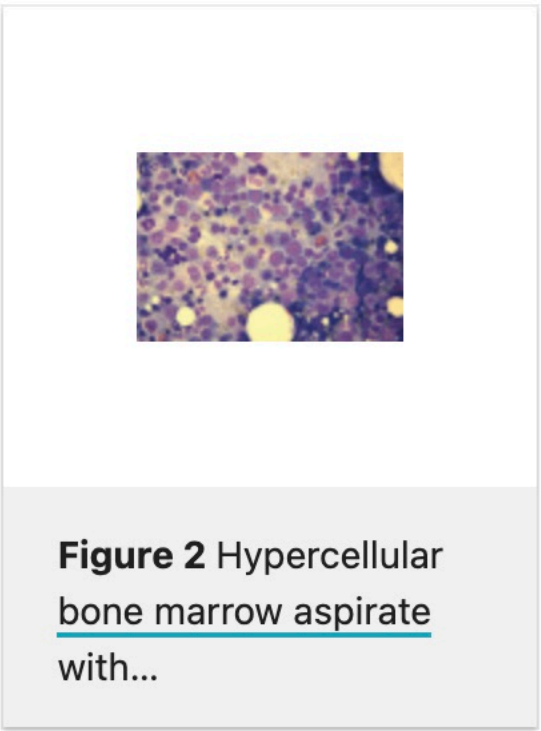
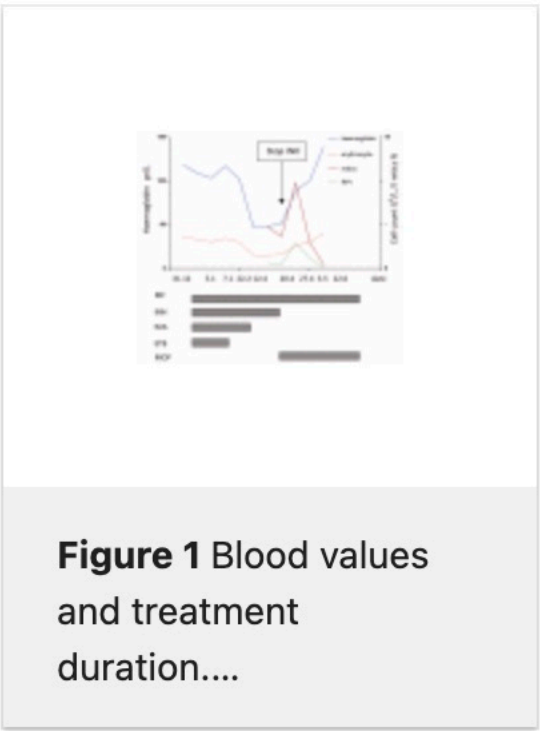
Abstract

Isoniazid induced sideroblastic anemia is a rare event. We report case of a 45 year old Caucasian women with development of severe anaemia 4 month after introduction of Isoniazid as part of Tuberculosis treatment. While haemoglobin fell to 47 g/L and erythrocyte count to 1.5 G/L, reticulocytes were very low (reticulocyte production index of 0.48), but bone marrow aspirate showed an accelerated erythropoiesis with ringsideroblasts. Anaemia rapidly resolved after cessation of Isoniazid. We postulate an Isoniazid induced inhibition of the δ -Amino-levulinat-synthase resulting in marked depletion of heam synthesis.

Keywords: isoniazid; sideroblastic anaemia; tuberculosis..

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Mark Question

A 47-year-old woman comes to the office due to an itchy rash. For the last 3 months, she has had a gradual onset of multiple pruritic lesions at her wrists, legs, and ankles. The patient tried treating the lesions with emollients and topical over-the-counter hydrocortisone but had no relief. Examination shows scattered, scaly, pink papules and plaques as shown in the [exhibit](#). Which of the following is the most likely diagnosis?

A. Actinic keratosis

ab

B. Atopic dermatitis

ab

C. Irritant contact dermatitis

ab

D. Lichen planus

ab

E. Psoriasis

ab

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 10 secs
Time Spent

- Chronic symptoms
- Formation of lesions at sites of trauma (Köbner phenomenon)
- Resolution within 2 yr (mucosal lesions may persist/recur)

(Choice B) Atopic dermatitis typically occurs in large patches at the flexural surfaces of the extremities, especially the elbows; it usually presents in childhood, and most patients have a history of other atopic disorders

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Mark Question

A 47-year-old woman comes to the office due to an itchy rash. For the last 3 months, she has had a gradual onset of multiple pruritic lesions at her wrists, legs, and ankles. The patient tried treating the lesions with emollients and topical over-the-counter hydrocortisone but had no relief. Examination shows scattered, scaly, pink papules and plaques as shown in the [exhibit](#). Which of the following is the most likely diagnosis?

A. Actinic keratosis (3%)

B. Atopic dermatitis (6%)

C. Irritant contact dermatitis (8%)

D. Lichen planus (68%)

E. Psoriasis (12%)

Incorrect

Correct answer D

68%

Answered correctly

10 secs

Time Spent

Natural history

- Chronic symptoms
- Formation of lesions at sites of trauma ([Köbner phenomenon](#))
- Resolution within 2 yr ([mucosal](#) lesions may persist/recur)

This patient has [lichen planus \(LP\)](#), an immune-mediated condition that typically presents with **pruritic, pink papules** and [plaques](#). LP lesions often have lacy, scaly, **white markings** ([Wickham striae](#)) and can form at sites of minor trauma ([Köbner phenomenon](#)). The skin lesions typically occur symmetrically on the flexural surfaces of the wrists and ankles but can also involve the nails, oral [mucous membranes](#), and genitalia.

The pathologic process in LP is characterized by a (CD8⁺) **T cell-mediated response** against cells along the [dermal-epidermal junction](#). Although the etiology is unknown, the risk may be increased in patients with [hepatitis C](#) and those taking certain medications (eg, [ACE inhibitors](#), [thiazide diuretics](#)). LP typically follows a chronic course with gradual, spontaneous remission within 2 years.

(Choice A) [Actinic keratosis](#) presents with small, [hyperkeratotic plaques](#) in sun-exposed areas (eg, forehead, cheeks). It is most common in patients age >60, and itching is atypical.

(Choice B) [Atopic dermatitis](#) typically occurs in large patches at the flexural surfaces of the extremities, especially the [elbows](#); it usually presents in childhood, and most patients have a history of other [atopic](#) disorders (eg, [allergic rhinitis](#), [asthma](#)).

(Choice C) [Irritant contact dermatitis](#) is due to repeated exposure to irritants (eg, detergents, solvents, oxidizing agents). It presents as [hyperkeratotic](#) patches with fissuring and scaling, most commonly on the hands.

(Choice E) [Psoriasis](#) is an immune-mediated inflammatory disease that usually presents with thick, [erythematous plaques](#) with silvery scales on the extensor surfaces of the knees and [elbows](#), scalp, or neck.

Educational objective:

[Lichen planus](#) is an immune-mediated condition that presents with pruritic, pink [papules](#) and [plaques](#), often with lacy, scaly, white markings ([Wickham striae](#)). The lesions typically occur on the flexural surfaces of the wrists and ankles but can also involve the nails, oral [mucous membranes](#), and genitalia.

References

[Update on lichen planus and its clinical variants.](#)

Pathology

Dermatology

[Lichen planus](#)

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☐

Mark Question

An 18-year-old man comes to the office due to 2 days of right testicular pain. The pain is constant and exacerbated by movement. Vital signs are within normal limits. On physical examination, there is swelling and tenderness to palpation localized only to the posterior and superior areas of the right testis. The left testis is normal. Cremasteric reflexes are intact bilaterally. Urinalysis shows numerous leukocytes but no bacteria. Which of the following factors most likely contributed to this patient's current condition?

☐ A. Inadequate childhood vaccination

☐ B. Increased gonadal venous pressure

☐ C. Lack of normal testicular fixation

☐ D. Unprotected sexual intercourse

☐ E. Urethral colonization by coliforms

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Kidney simulator

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Mark Question

An 18-year-old man comes to the office due to 2 days of right testicular pain. The pain is constant and exacerbated by movement. Vital signs are within normal limits. On physical examination, there is swelling and tenderness to palpation localized only to the posterior and superior areas of the right testis. The left testis is normal. Cremasteric reflexes are intact bilaterally. Urinalysis shows numerous leukocytes but no bacteria. Which of the following factors most likely contributed to this patient's current condition?

A. Inadequate childhood vaccination (17%)

B. Increased gonadal venous pressure (11%)

C. Lack of normal testicular fixation (11%)

D. Unprotected sexual intercourse (53%)

E. Urethral colonization by coliforms (5%)

Incorrect

Correct answer D

53%

Answered correctly

05 secs

Time Spent

Explanation

Acute epididymitis	
Epidemiology	- Age <35: sexually transmitted (chlamydia, gonorrhea) - Age >35: <u>bladder outlet obstruction</u> (coliform bacteria)
Symptoms	- Unilateral testicular pain - Epididymal edema <u>Dysuria</u>, frequency (with coliform infection)
Diagnosis	<u>NAAT</u> for <u>chlamydia</u> and <u>gonorrhea</u> <u>Urinalysis/culture</u>
<u>NAAT</u> = nucleic acid amplification test; <u>STI</u> = sexually transmitted infection.	

This patient's acute testicular pain, **posterior testicle tenderness**, and pyuria raises suspicion for **acute epididymitis**. Most cases occur when genitourinary pathogens travel in a retrograde fashion from the urethra via the ejaculatory duct to the vas deferens. The most likely underlying organism can often be inferred based upon the age of the patient:

- Young men (age <35) usually develop acute epididymitis due to ***Chlamydia trachomatis*** or ***Neisseria gonorrhoeae***, sexually transmitted pathogens acquired during **unprotected sexual intercourse**. Although these organisms often cause asymptomatic urethritis (no dysuria), **pyuria** is typically seen on urinalysis. Nucleic acid amplification testing is required for diagnosis because urine culture is generally negative.
- Older men (age >35) are less likely to have sexually transmitted infections due to fewer sexual partners and increased rates of monogamy. Acute epididymitis in these patients is generally due to gram-negative colonic flora (eg, *Escherichia coli*), which contaminate the urethra, ascend into the urinary system, and subsequently invade the ejaculatory duct and vas deferens. Most cases arise in the setting of obstruction (eg, benign prostatic hypertrophy) and are characterized by urinary tract symptoms (dysuria, urinary frequency), bacteruria on urinalysis, and positive urine culture (**CH**).

(Choice A) Mumps can occur in those who did not receive appropriate childhood vaccination.

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05 secs

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Although these organisms often cause asymptomatic urethritis (no dysuria), pyuria is typically seen on urinalysis. Nucleic acid amplification testing is required for diagnosis because urine culture is generally negative.

Older men (age >35) are less likely to have sexually transmitted infections due to fewer sexual partners and increased rates of monogamy. Acute epididymitis in these patients is generally due to gram-negative colonic flora (eg, Escherichia coli), which contaminate the urethra, ascend into the urinary system, and subsequently invade the ejaculatory duct and vas deferens. Most cases arise in the setting of urethral obstruction (eg, benign prostatic hypertrophy) and are characterized by urinary tract symptoms (eg, dysuria, urinary frequency), bacteruria on urinalysis, and positive urine culture (Choice E).

(Choice A) Mumps can occur in those who did not receive appropriate childhood vaccinations and usually presents with a nonspecific prodrome (fever, malaise, myalgias) followed by parotitis. Complications include orchitis, with high fever and severe diffuse testicular pain, which are not seen in this patient.

(Choice B) Varicocele is characterized by distension of the pampiniform plexus due to elevated gonadal venous pressure. It presents as a soft scrotal mass that feels like a "bag of worms" on palpation. Pain is mild or absent.

(Choice C) Testicular torsion is the result of inadequate fixation of the lower pole of the testis to the tunica vaginalis. It is usually characterized by sudden-onset, severe, unilateral testicular pain; nausea; a high-riding testis; and loss of the cremasteric reflex (elevation of testis on pinching of the skin at the upper thigh).

Educational objective:

Epididymitis presents with acute testicular pain, tenderness, and pyuria. It is caused by retrograde passage of organisms from the urethra into the ejaculatory duct and vas deferens. The microbiology is largely influenced by patient age: epididymitis in young men is usually due to sexually acquired infections (eg, Chlamydia trachomatis, Neisseria gonorrhoeae), whereas in older men (age >35) it is usually due to gram-negative colonic flora.

References

Epididymitis: an overview.